Table 4-5. *Bitwise operators*

&	Bitwise AND – returns a value that has bits set (to *1*) where both operands have the same bits set
|	Bitwise OR – returns a value that has bits set where either operand also has a bit set
<<	Shift left all bits in the left-hand operand the number of places specified by the right-hand (each step multiplies by two)
>>	Shift right all bits in the left-hand operand the number of places specified by the right-hand (each step divides by two

You can probably work out the gotchas for yourself: bitwise & and logical &&, bitwise << and comparison <, and so on.

Errors and Exceptions

An important aspect of control in any code you write is dealing with any problems that might arise during execution. With a little forethought and care, any issue that might arise (perhaps even through no fault of your own) that might otherwise result in meaningless garbage can be transformed into useful information, or even something still useful to an end user. PHP has two principal mechanisms for when things go wrong: a set of internal error conditions, which if met will result in the emission of an error message, which has a variety of effects according to configuration, and an object-oriented exception model that can be used to gracefully communicate and handle runtime problems between components.

PHP errors can prevent the program from continuing (in the worst cases) or merely produce a message and carry on. Exceptions, on the other hand, are intended to interrupt the flow of the program and only prevent continuation if you choose to do so (or forget to write the code to deal with them). The language constructs for handling exceptions aren't strictly counted as control structures in other resources, but since they can impact the flow of logic, I've opted to include them in this chapter. PHP makes use of exceptions in all of its OOP APIs and in many other places too (there was a big push in version 7 to switch to exceptions over errors). The language also provides the function `trigger_error()`, which allows you to deliberately emit an error, though the exception model is generally preferred these days.